

WEEK 1 – POWER BI PROJECT REPORT

Data Import, Cleaning and Transformation of Superstore Sales Dataset

1. INTRODUCTION

Power BI is a data visualization and business intelligence tool used to analyze data and create reports and dashboards. In this Week 1 task, the Superstore Sales Dataset was imported into Microsoft Power BI and prepared using Power Query Editor.

The main focus of this task was to understand the dataset, check data quality, correct data types, clean text values, check blank and error values, rename columns, and create useful date attributes from the Order Date column. These steps help prepare the dataset for further analysis and visualization.

2. OBJECTIVES

The main objectives of Week 1 are:

- To import the Superstore Sales Dataset into Power BI.
- To understand the structure of the dataset.
- To inspect the dataset using Power Query Editor.
- To check and correct data types.
- To check blank, empty, null and error values.
- To clean text data using Trim and Clean.
- To check record uniqueness.
- To rename columns where required.
- To perform column transformations.
- To create Year, Quarter, Month, Day and Day of Week from Order Date.
- To validate the cleaned dataset.
- To load the final dataset into the Power BI data model.

3. TOOLS AND DATASET

Tool Used

- Microsoft Power BI Desktop
- Data Preparation Tool
- Power Query Editor
- Dataset Name
- Superstore Sales Dataset

File Type

CSV

Power Query Editor was used to import, inspect, clean and transform the dataset before loading it into Power BI.

4. DATASET OVERVIEW

The Superstore Sales Dataset contains information about sales transactions in a retail business.

Major information includes:

Order Information

- Row ID
- Order ID
- Order Date
- Ship Date
- Ship Mode

Customer Information

- Customer ID
- Customer Name
- Segment

Product Information

- Category
- Sub-Category
- Product Name

Sales Information

- Sales
- Quantity
- Discount
- Profit

The dataset provides the information required for studying sales and profitability patterns.

5. DATA IMPORT

The Superstore Sales Dataset was imported into Power BI using the Get Data → Text/CSV option. After selecting the CSV file, the dataset was previewed and Transform Data was selected to open it in Power Query Editor.

Steps followed:

- Opened Power BI Desktop.
- Selected Home.
- Selected Get Data.
- Selected Text/CSV.
- Selected the Superstore Sales Dataset.
- Clicked Open.
- Checked the data preview.
- Selected Transform Data.
- Opened the dataset in Power Query Editor.

Screenshot – Data Import / Power Query

Row ID	Order ID	Order Month	Ship Month	Ship Mode
1	US-2023-99800	January	January	STANDARD CLASS
2	US-2023-112335	January	January	STANDARD CLASS
3	US-2023-112525	January	January	STANDARD CLASS
4	US-2023-112335	January	January	STANDARD CLASS
5	US-2023-141632	January	January	STANDARD CLASS
6	US-2023-152194	January	January	STANDARD CLASS
7	US-2023-157199	January	January	STANDARD CLASS
8	US-2023-996054	January	January	FIRST CLASS
9	US-2023-157199	January	January	STANDARD CLASS
10	US-2023-157199	January	January	STANDARD CLASS
11	US-2023-157199	January	January	STANDARD CLASS
12	US-2023-133853	January	January	SECOND CLASS

6. NAVIGATING POWER BI VIEWS

Power BI provides three main views:

Report View

Report View is used to create charts, visualizations and reports.

Data View

Data View is used to inspect the rows and columns of the dataset.

Model View

Model View is used to view and manage tables and relationships in the data model.

7. CHECKING AND CHANGING DATA TYPES

After importing the dataset, the data type of each required column was checked.

Correct data types are important for accurate analysis and calculations.

Data types used:

Text columns → Text

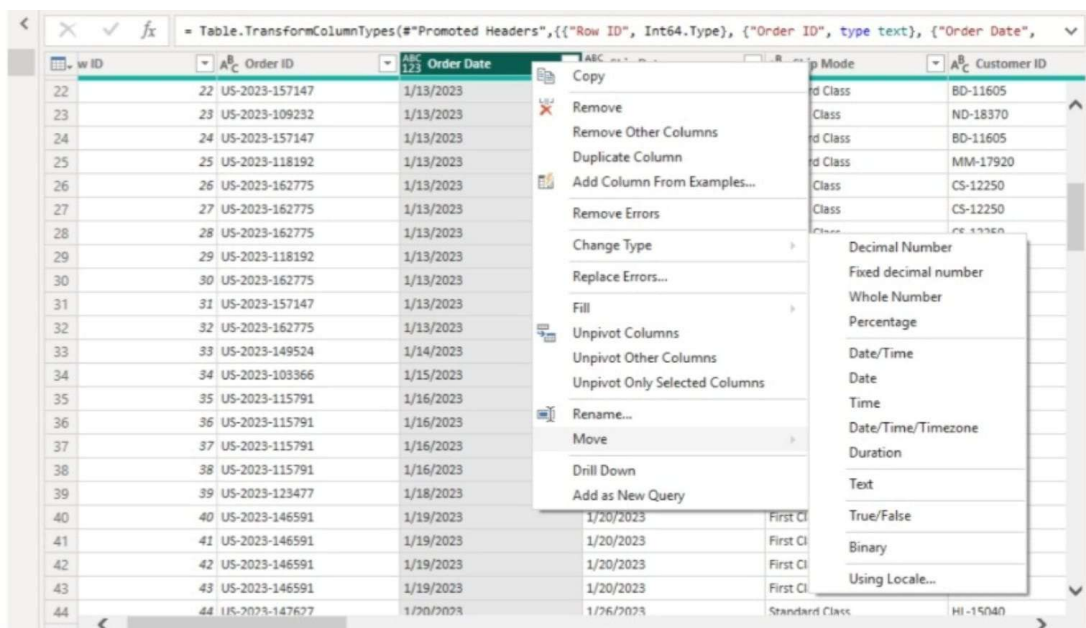
Quantity-related columns → Whole Number

Sales and Profit → Decimal Number

Order Date → Date

The Order ID column was also checked to ensure that it was treated as text rather than as a date or number.

Screenshot – Data Type Correction



The screenshot shows the Change Type option in Power Query, including Text, Whole Number, Decimal Number and Date.

8. CLEANING TEXT DATA USING TRIM AND CLEAN

Categorical columns were cleaned to improve consistency.

The main categorical columns cleaned were:

- Category
- Sub-Category
- Segment

Trim

Trim was used to remove unnecessary spaces from text values.

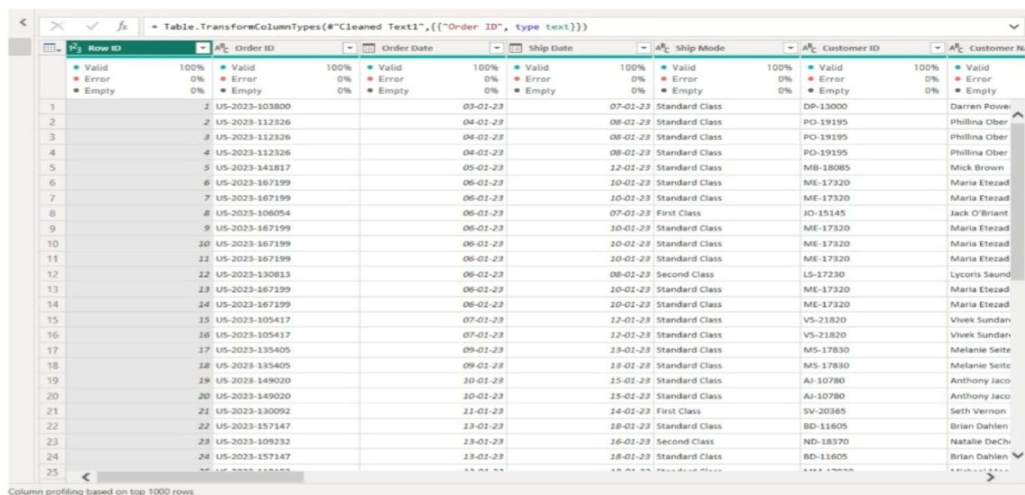
Clean

Clean was used to remove unwanted non-printable characters.

Steps:

1. Selected the required column.
2. Selected Transform.
3. Selected Format.
4. Selected Trim.
5. Checked the cleaned values.
6. Selected Transform → Format → Clean.
7. Checked the data again.
8. Repeated the process for the other categorical columns.

Screenshot – Data Cleaning



Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name
1	US-2023-103800	03-01-23	07-01-23	Standard Class	DP-13000	Darren Power
2	US-2023-112326	04-01-23	08-01-23	Standard Class	PO-19195	Phillina Ober
3	US-2023-112326	04-01-23	08-01-23	Standard Class	PO-19195	Phillina Ober
4	US-2023-112326	04-01-23	08-01-23	Standard Class	PO-19195	Phillina Ober
5	US-2023-141817	05-01-23	12-01-23	Standard Class	MB-18085	Mick Brown
6	US-2023-167199	06-01-23	10-01-23	Standard Class	ME-17320	Maria Etezad
7	US-2023-167199	06-01-23	10-01-23	Standard Class	ME-17320	Maria Etezad
8	US-2023-106054	06-01-23	07-01-23	First Class	JO-15145	Jack O'Briant
9	US-2023-167199	06-01-23	10-01-23	Standard Class	ME-17320	Maria Etezad
10	US-2023-167199	06-01-23	10-01-23	Standard Class	ME-17320	Maria Etezad
11	US-2023-167199	06-01-23	10-01-23	Standard Class	ME-17320	Maria Etezad
12	US-2023-130813	06-01-23	08-01-23	Second Class	LS-17230	Lyons Saund
13	US-2023-167199	06-01-23	10-01-23	Standard Class	ME-17320	Maria Etezad
14	US-2023-167199	06-01-23	10-01-23	Standard Class	ME-17320	Maria Etezad
15	US-2023-105417	07-01-23	12-01-23	Standard Class	VS-21820	Vivek Sundan
16	US-2023-105417	07-01-23	12-01-23	Standard Class	VS-21820	Vivek Sundan
17	US-2023-135405	09-01-23	13-01-23	Standard Class	MS-17830	Melanie Sette
18	US-2023-135405	09-01-23	13-01-23	Standard Class	MS-17830	Melanie Sette
19	US-2023-149020	10-01-23	15-01-23	Standard Class	AJ-10780	Anthony Jaco
20	US-2023-149020	10-01-23	15-01-23	Standard Class	AJ-10780	Anthony Jaco
21	US-2023-130092	11-01-23	14-01-23	First Class	SV-20365	Seth Vernon
22	US-2023-157147	13-01-23	18-01-23	Standard Class	BD-11605	Brian Dahlen
23	US-2023-109232	13-01-23	16-01-23	Second Class	ND-18370	Natalie DeCh
24	US-2023-157147	13-01-23	18-01-23	Standard Class	BD-11605	Brian Dahlen
25	US-2023-109232	13-01-23	16-01-23	Second Class	ND-18370	Natalie DeCh

The screenshot shows the cleaned dataset with Valid, Error and Empty values displayed through column profiling.

9. CHECKING BLANK, EMPTY, NULL AND ERROR VALUES

After cleaning the text data, the dataset was checked for blank, empty, null and error values.

Steps followed:

1. Selected the required column.
2. Clicked the Filter option.
3. Checked the values in the filter list.
4. Looked for blank or empty values.
5. Checked for null values.
6. Checked for Error values.
7. Reviewed affected columns.
8. Handled unnecessary values where required.
9. Rechecked the columns after cleaning.

The column profiling shown in the screenshot helps identify whether the values are Valid, Error or Empty.

10. COLUMN TRANSFORMATION

Column transformations were performed to organize the dataset and make it easier to understand.

Rename Column

Columns were renamed where necessary to provide clear and meaningful names.

Steps:

1. Selected the required column.
2. Right-clicked the column.
3. Selected Rename.
4. Entered the required column name.
5. Checked the renamed column.
- 6.

11. CREATING DATE ATTRIBUTES

The Order Date column was used to create useful date-related columns.

The following attributes were created:

- Year
- Quarter
- Month
- Day
- Day of Week

These columns are useful for future time-based sales analysis.

Year

Order Date → Add Column → Date → Year

Quarter

Order Date → Add Column → Date → Quarter → Quarter of Year

Month

Order Date → Add Column → Date → Month

Day

Order Date → Add Column → Date → Day

Day of Week

Order Date → Add Column → Date → Day → Name of Day

12. FINAL DATA VALIDATION

After completing the cleaning and transformation process, the dataset was checked again.

Final checks included:

- Data types
- Order Date
- Year
- Quarter
- Month
- Day
- Day of Week
- Blank and empty values
- Null values
- Error values
- Cleaned text data

- Renamed columns
- Transformed columns
- Final dataset structure

13. CLOSE AND APPLY

After completing all the required transformations:

1. Checked the final dataset in Power Query Editor.
2. Went to the Home tab.
3. Clicked Close & Apply.
4. Waited for Power BI to apply the changes.
5. The cleaned dataset was loaded into the Power BI data model.

14. FINAL RESULT

The Superstore Sales Dataset was successfully imported and prepared using Power BI and Power Query Editor.

The data types were checked and corrected. Text data was standardized using Trim and Clean. Blank, empty, null and error values were checked. Column transformations were performed, and useful date attributes were created from Order Date. Finally, the cleaned dataset was validated and loaded into the Power BI data model.

15. CONCLUSION

In Week 1, I learned the basic process of preparing data using Power BI and Power Query Editor. I successfully imported the Superstore Sales Dataset, checked its quality, corrected data types and cleaned the required columns. I also created useful date attributes from Order Date. This prepared the dataset for further analysis and dashboard creation in the upcoming stages.